Asking Retraction Watch Twice: Retraction Data Revise Their Own Past

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Abstract

Since Crossref opened the Retraction Watch database in 2023, it has been republished daily, and studies cite it as if it were one fixed dataset. We replayed 39 snapshots spanning its public history (2024-09-02 to 2026-10-16) and asked each the same four questions about retractions dated 2023 or earlier, so that any change is the database revising the past. Over two years such retractions grew 9.1%, those citing "Paper Mill" grew 80.6%, the median time to retraction rose from 379 to 420 days, and records citing "Fake Peer Review" fell from 4,654 to zero when the label was renamed. Nearly all the growth is late cataloguing: 4,482 records added after the first snapshot, 3,921 of them Hindawi retractions. Joining Retraction Watch with OpenAlex, PubMed, and PubMed Central, all eligible registries agree on 66.1% of 89,706 flagged papers, and 39.2% of the works OpenAlex marks as retracted are retraction notices, which its ingest rule flags along with the papers they retract. A blind hand audit of papers that only one registry flags estimates that 97.4% (93.7% to 100%) were retracted, so most disagreement is a registry missing a retraction. We recommend citing and archiving the snapshot a study uses.

Keywords: retraction; Retraction Watch; OpenAlex; research integrity; reproducibility; metascience

1. Introduction

A study that reports "N retractions for paper mill activity through 2023" reports a property of the database on the day it was queried, not a property of the literature. That would not matter if retraction databases only ever appended new notices. They do not. Records are added years after the retraction, reasons are renamed in bulk, and dates and identifiers are corrected.

We ask how different the answers are when two researchers put the same question about the past to Retraction Watch on different days, and why. That needs the database's history as well as its current state. Crossref publishes the Retraction Watch database as a CSV in a public GitLab repository, regenerated daily, and every regeneration is a commit. The history therefore exists, and the maintainers log taxonomy changes, but no study we found reads it as a time series of records. The closest work compares two time points across indexes (Salami et al., 2024b) or tracks Crossref DOIs that lost their retracted flag, and names longitudinal study of data quality as future work (Si et al., 2026).

We make four contributions.

1. A replay of 39 Retraction Watch snapshots covering the whole public history, keeping every version of every record, so the database can be queried as of any snapshot date.
2. A fixed set of questions asked of every snapshot, showing how much and through which mechanisms the answers move (Section 4.1).

3. A four registry comparison (Retraction Watch, OpenAlex, PubMed, PubMed Central) on current data, which replicates the disagreement that earlier studies found across other indexes, and independently measures how often OpenAlex's retraction flag marks the notice rather than the paper (Sections 4.2 and 4.3).
4. A blind hand audit of papers that only one registry flags, judged against publishers rather than against the union of the registries, with its codebook committed before labelling (Section 4.4).

2. Related work

Indexes disagree about what is retracted. This is well established. Schneider et al. (2023) compared Crossref, Retraction Watch, Scopus, and Web of Science and found only 3% of 49,924 DOIs flagged by all four. Salami et al. (2024a, corrected 2026) extended the comparison to 11 sources and found at least one disagreement on 92.64% of 83,317 items. Ortega and Delgado-Quirós (2024) compared coverage across seven databases including OpenAlex and PubMed, and Sebo and Sebo (2025) compared Retraction Watch, PubMed, and Web of Science on 878 retracted publications in 131 high impact medical journals. We do not claim disagreement as a finding. Section 4.2 replicates it with OpenAlex and PMC, which the 11 source study did not include.

Notices indexed as retracted papers. Hand checks by the same group found that a retraction index often holds the notice rather than the paper: 59 of 201 DOIs sampled from those on which the four sources disagreed in Schneider et al. (2023), and 661 of 925 sampled Crossref DOIs (71.46%) in Si et al. (2026). For OpenAlex, Hauschke and Nazarovets (2025) reported a different failure: between 2023-12-22 and 2024-03-19 OpenAlex set is_retracted for any work carrying an update relation, such as a correction or an expression of concern, and OpenAlex reported fixing about 2,300 works. A company report (Tesseract Semantics, 2026), not peer reviewed, states that OpenAlex "flags 94.5 per cent of Retraction Watch notice DOIs as retracted". Section 4.3 measures the same thing independently.

Retraction data as a moving target. Si et al. (2026) found that between April 2023 and July 2024 Crossref removed the retracted flag from 208 DOIs, none of which was in fact retracted, and concluded that "retraction metadata are dynamic rather than static". Their study covers Crossref only, and they name longitudinal assessment of data quality as future work. Salami et al. (2024b) compared union lists built in April 2023 and July 2024. Their own 11 source dataset was later corrected because it had included Retraction Watch records whose nature is not a retraction (expressions of concern, corrections, reinstatements) (Salami and McCumber, 2026), so even specialists can misread the database's structure. Retraction Watch documents its taxonomy changes in a dated changelog (Retraction Watch, n.d.). We found no study that compares Retraction Watch releases record by record, or that measures what those changes do to the answers a study gets.

Citation after retraction. Retracted papers keep being cited, mostly without acknowledgement of the retraction (Bar-Ilan and Halevi, 2017; Schneider et al., 2020); Hsiao and Schneider (2021) found that only 722 of 13,252 (5.4%) post retraction citation contexts in the PubMed Central Open Access Subset acknowledged the retraction. Our citation counts (Section 4.5) measure volume at year granularity and do not classify contexts.

3. Data and methods

3.1 Sources

All four sources were fetched on 2026-10-19 from public endpoints: the Retraction Watch Database (2026), OpenAlex (Priem et al., 2022), PubMed, and the PMC Article Datasets (National Library of Medicine, 2026).

Source	What we fetched
Retraction Watch (Crossref GitLab)	39 snapshots, 2024-09-02 to 2026-10-16; 55,804 records in the first, 73,283 in the last
OpenAlex	136,873 of 136,970 works with `is_retracted: true`
PubMed	35,169 PMIDs with publication type "Retracted Publication"
PubMed Central	metadata for 43,616 articles the other registries mention

The GitLab history begins on 2024-09-02 and held 518 daily snapshots when we listed it on 2026-10-26. We replayed the first snapshot of each month through 2026-09-01 and every snapshot from 2026-09-29 on.

3.2 Snapshot replay

Each snapshot is landed unchanged and replayed in date order. A record is keyed by its Retraction Watch Record ID. When a record's content changes between consecutive snapshots, its current version is closed and a new one opened, so every version carries a validity interval (`valid_from` inclusive, `valid_to` exclusive). The view of the database as of a date d is the set of versions valid at d. Changes that only alter formatting (whitespace, a dropped + prefix on every reason, capitalization) are classified separately from semantic edits, so a reformatting of the whole file does not read as 143,447 edits.

One snapshot, 2026-10-15, arrived in cp1252 rather than UTF-8 and had lost characters in 5,520 records. It is quarantined rather than replayed as edits.

Validation against the maintainers' changelog. Retraction Watch logs its taxonomy and format changes with dates (Retraction Watch, n.d.). Every change it logs inside our window appears in the first replayed snapshot after its logged date, classified the right way:

Logged	Change	First replayed snapshot	What the replay recorded
2024-09-23	"Withdrawal" renamed "Removed"	2024-10-01	relabel, 3,798 records
2025-02-28	about a hundred reason renames	2025-03-03	relabels, 28,223 updated records
2025-03-06	"Unreliable Results" renamed	2025-04-01	relabel, 16,379 records
2025-08-21	"+" prefix removed from reasons	2025-09-02	format only, 65,898 records
2025-11-25	"Concerns/Issues with Peer Review" renamed	2025-12-01	relabel, 10,557 records
2025-12-12	"Fake Peer Review" renamed	2026-01-02	relabel, 10,426 records

The replay recovered every documented change without being told about it, which gives us grounds to treat the undocumented changes (late records, DOI and date edits, deletions) as real.

3.3 The same question on every snapshot

We fixed four questions before computing any of them, and report all four:

1. How many retractions are dated 2023 or earlier?
2. How many of those cite the reason "Paper Mill"?
3. How many of those cite the reason "Fake Peer Review"?
4. What is their median time from publication to retraction?

Restricting every question to retractions dated 2023 or earlier means a correct, append only database would give the same answer on every snapshot after early 2024 (allowing for notices dated late in 2023 that take a few weeks to be entered). Any larger change is the database revising the past. Reason matching is case insensitive substring matching on the reason field.

3.4 Cross registry identity and agreement

Papers are matched across sources on DOI, PMID, and PMCID, using one hop of the PMC identifier map. A paper is in the registry if Retraction Watch holds a record of it or any source flags it as retracted. Retraction and withdrawal notices are excluded before matching: OpenAlex works of type retraction, and any work whose DOI or PMID Retraction Watch records as a notice, unless Retraction Watch also records it as the retracted paper's own identifier. IEEE and some other publishers stamp the notice onto the paper's page, so the two share one DOI, and that DOI is the paper.

A registry is eligible for a paper only if it could have flagged it: PubMed for papers with a PMID, PMC for papers with a linked open copy, OpenAlex and Retraction Watch for every paper. "All eligible agree" means every eligible registry flags the paper.

3.5 Hand audit

Agreement against the union of registries says nothing about whether the union is right. We drew a sample from the 36,401 papers that exactly one registry flags, stratified by that registry with equal allocation: all 22 papers that only PubMed flags, and 89 each from those that only Retraction Watch (20,259) or only OpenAlex (16,120) flags. No paper is flagged by PMC alone. The draw is seeded and reproducible.

Labels follow a codebook (`docs/audit-codebook.md`) committed, with the blank sheet, before any label was entered (commit 772bdad, 2026-10-26). Labels are assigned from the publisher's version of record or the publisher's own notice, never from the four registries under test. The sheet withholds the flagging registry and strips leading "RETRACTED:" and "WITHDRAWN:" markers from titles (86 of 200), which would reveal both the source and the answer. Blinding is partial: 27 papers have no DOI, PMID, or PMCID, and nearly all of those are Retraction Watch records; and 19 of the 22 PubMed only papers have a PMID but no DOI. A labeller could infer the source from which links exist.

After the first labelling was complete and scored, a second labeller labelled 40 of the 200 rows, drawn with a fixed seed from the blank sheet. The second labeller worked alone from the same codebook and never saw the first labels, the flagging registry, or the scores (codebook change log, 2026-10-28).

Precision per stratum is the share labelled retracted among papers not labelled `cannot_determine`, with Wilson 95% intervals. The frame estimate weights strata by population with a finite population correction, so the nearly complete PubMed stratum adds little sampling variance. The frame was later corrected (see Correction): it now holds 22,510 papers, and 60 sampled papers, all from the Retraction Watch stratum, left it because another registry in fact flags them. The sampled papers still in each stratum remain a random sample of it, so the estimates in Section 4.4 use them with the corrected stratum sizes: 6,345 Retraction Watch, 16,141 OpenAlex, and 24 PubMed.

3.6 Notice links in PMC

To see how PMC keeps notices apart from papers, we read the JATS XML of every retraction notice in PMC's open data, fetched on 2026-10-29. The frame is the 34,110 records PubMed types as retraction notices plus Retraction Watch's retraction notice identifiers, mapped through PMC-ids.csv to 22,098 PMCIDs; 18,574 are in the open data and 18,296 of those are typed `retraction`. A notice counts as linked when it carries a `related-article` element of type `retracted-article` with an identifier. The XML is PMC's processed copy, so this measures PMC's record, not what the publisher deposited. PubMed's term for notices is now `"retraction notice"[pt]`; the older `"retraction of publication"[pt]` returned no records that day.

4. Results

4.1 Retraction Watch revises its past

Table 1 gives the four answers on the first and last snapshots; Figure 1 shows every snapshot.

Table 1. The same four questions, asked 25 months apart.

Question	2024-09-02	2026-10-16	Change
Retractions dated 2023 or earlier	48,789	53,227	+9.1%
Of those, citing "Paper Mill"	4,868	8,790	+80.6%
Of those, citing "Fake Peer Review"	4,654	0	−100%
Median days, publication to retraction	379	420	+10.8%

Three mechanisms account for the movement.

Late cataloguing. 4,482 records with a retraction date in 2023 or earlier were added after the first snapshot. 4,406 of them carry a Record ID above the first snapshot's highest, so they are new catalogue entries rather than rows the first file happened to omit. 4,315 are dated 2023, and Hindawi accounts for 3,921. Retractions dated 2023 alone went from 9,030 to 13,337: a study of 2023 run in September 2024 would have seen about two thirds of what the database now holds. The late records also explain the other movements: 3,924 of them cite "Paper Mill" (against a net increase of 3,922), and their median lag is 561 days, which pulls the cohort median up. Most arrived between October 2024 and March 2025 (Figure 2), but they were still arriving in October 2026.

Relabelling. The reason taxonomy was renamed in bulk several times. On 2026-01-02 "Fake Peer Review" became "Compromised Peer Review" on 10,426 records; on 2025-04-01 "Unreliable Results" became "Unreliable Results and/or Conclusions" on 16,379. A query written against the old label silently returns zero. Of the 73,334 records ever seen, 50,518 (68.9%) were edited after they first appeared, mostly through these relabels.

Edits and deletions. 3,840 records (5.2%) changed in something other than reasons: 96 changed their original paper DOI, 54 their retraction date, and 27 their nature (for example, from expression of concern to retraction). 54 records were deleted and 3 reappeared. Within the 2023 cohort these are small: 22 records deleted and 22 moved out by a date or nature edit.

4.2 The registries disagree

89,706 papers are flagged retracted, or disputed, by at least one registry. All eligible registries agree on 59,272 (66.1%), and exactly one flags 22,510 (25.1%).

Table 2. Each registry against the union. The union is not ground truth; Section 4.4 measures it.

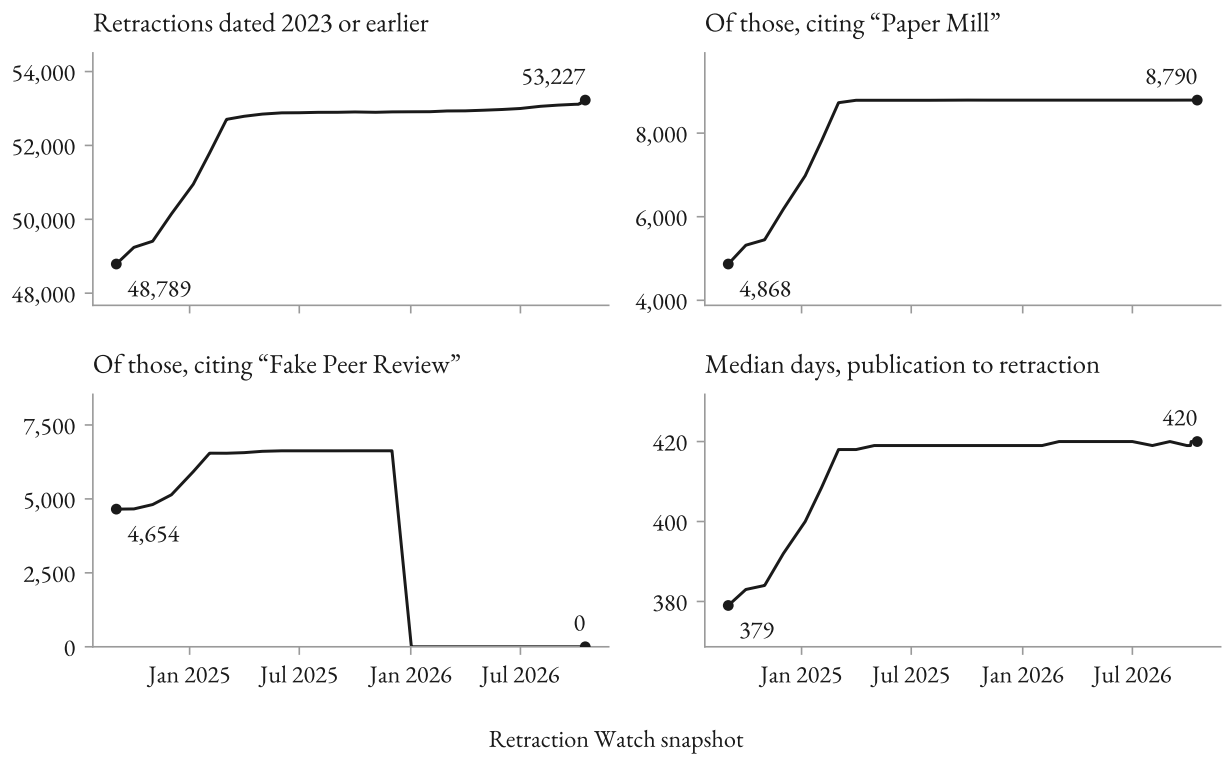


Figure 1. Each panel asks one question of all 39 snapshots.

Registry	Eligible papers	Flags	Share of the union
OpenAlex	84,187	82,866	98.4%
PMC	19,867	18,880	95.0%
PubMed	37,528	34,561	92.1%
Retraction Watch	89,706	67,672	75.4%

These are not directly comparable with Salami et al. (2024a), who measure agreement over the union of 11 sources without conditioning on whether a source could have held a paper. The one registry both studies include behaves alike: in Salami et al. (2024a) PubMed had the highest agreement score, indexing as retracted 22,766 (87.69%) of the 25,962 union list items it covers, and here it covers 92.1% of the eligible union. Retraction Watch sits lowest here (75.4%) because the union includes OpenAlex, whose flag reaches many papers no other registry marks; Section 4.4 asks how many of those are real.

By publisher. Agreement depends more on the publisher than on the registry (Table 3). For the 12,468 Hindawi papers, the largest group, all eligible registries agree on 94.4%, and for IOS Press on 90.7%. For EDP Sciences only 49.1% agree and 50.9% are flagged by a single registry, and IEEE, Springer, Elsevier, and Taylor and Francis sit between 60.2% and 67.9%. A study's coverage therefore depends on which publishers dominate its sample, not only on which registry it uses.

Table 3. Agreement for the ten largest DOI prefixes. The publisher is the name Retraction Watch most often gives the prefix.

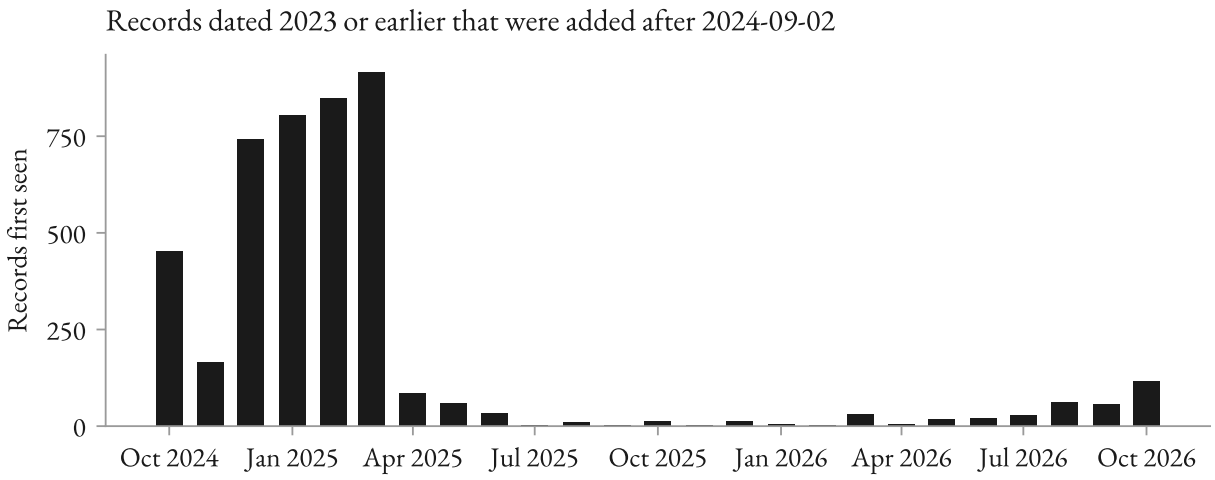


Figure 2. When the late records first appeared.

Publisher (prefix)	Papers	All eligible agree	One registry only
Hindawi (10.1155)	12,468	94.4%	2.2%
Elsevier (10.1016)	11,821	60.2%	23.8%
IEEE (10.1109)	11,439	67.9%	32.1%
Springer (10.1007)	9,389	66.7%	26.0%
Wiley (10.1002)	2,820	64.2%	8.5%
Taylor and Francis (10.1080)	2,380	60.3%	24.0%
Wiley (10.1111)	1,923	62.6%	12.6%
IOS Press (10.3233)	1,892	90.7%	6.8%
SAGE (10.1177)	1,608	72.8%	13.7%
EDP Sciences (10.1051)	1,571	49.1%	50.9%

4.3 Two in five of OpenAlex's retracted works are notices

Of the 136,873 OpenAlex works with `is_retracted: true`, 53,618 (39.2%) are retraction notices: 35,976 have type retraction, and 17,642 more carry a DOI or PMID that Retraction Watch records as a notice and not also as a paper. Counted as papers, they would add about 60% to the registry. PMC handles the same distinction well: of the 17,533 notice articles it holds, 55 carry its retracted flag.

From the notice's side the picture is the same. Of the 44,651 Retraction Watch retraction notice DOIs that are not also some paper's DOI, OpenAlex flags 41,460 (92.9%) as retracted. This independently reproduces, within two points, the 94.5% in the Tesseract Semantics (2026) report, on data fetched two months later.

OpenAlex's public ingest code shows why. It sets is_retracted when either a work's Crossref update-to relation or its updated-by relation carries a retraction label (OpenAlex, 2026). A retraction notice is the record that carries update-to, pointing at the paper it retracts, so the rule flags the notice along with the paper.

PMC shows that the distinction can be kept. Its flag follows a structured link: a notice names the paper it retracts in a related-article element of type retracted-article, and PMC marks that paper. Of the 18,296 records PMC types as retraction notices, 18,267 (99.8%) carry the link, and one carries the retracted flag itself. The 55 counted above were matched by identifier, a rule that also catches papers sharing an identifier with a notice. Every paper PMC holds

that both the notice's link and PubMed's "Retraction of" entry name, 17,398 of 17,398, is flagged, and PMC and PubMed name different papers for 4 notices. The two rules differ in direction: PMC flags the target of a notice's link, while OpenAlex flags the works at both ends of it.

This differs from the failure Hauschke and Nazarovets (2025) reported, which flagged corrected papers and ran for three months in early 2024. This one flags the notice that announces a retraction, is current, and affects nine in ten notices.

4.4 How many single registry flags are real

Of the 200 sampled papers, 183 could be decided from the publisher's page. The other 17 could not: 14 of them are papers only Retraction Watch flags, and 12 of those 14 have no DOI, PMID, or PMCID to follow. Across all 200, the labels were retracted 173 times, notice 6, not_retracted 3, and concern once. After the correction, 140 of the 200 remain in the frame.

Table 4. Precision of a flag that no other registry shares: the share the publisher actually retracted, among papers that could be decided.

Flagged only by	Papers in frame	Sampled, still in frame	Decided	Precision (95% CI)	Notices
OpenAlex	16,141	89	86	98.8% (93.7% to 99.8%)	1
PubMed	24	22	22	86.4% (66.7% to 95.2%)	1
Retraction Watch	6,345	29	16	93.8% (71.7% to 98.9%)	1

Weighted by stratum size, with a finite population correction, 97.4% (93.7% to 100%) of the 22,510 single registry flags are real retractions. Most disagreement, then, comes from one registry holding a retraction the others miss; few single flags point at papers that were never retracted. Treated as a measure of coverage, the union in Table 2 is close to right.

Three cautions apply. The OpenAlex figure is precision after the notice filter of Section 3.4, not of the raw is_retra cted flag, which Section 4.3 shows is two fifths notices. Papers flagged only by OpenAlex may still be in Retraction Watch under a record with no DOI or PMID, which one hop identity resolution cannot match. And the Retraction Watch stratum rests on 16 decided papers: counting all 13 undecided ones as not retracted, the worst case, it would be 15 of 29.

Notices counted as papers make up 3 of the 140 papers still in the frame, one in each stratum, the same failure as Section 4.3 at a smaller scale.

The two labellers agree on 37 of the 40 rows they share (92.5%, Cohen's kappa 0.80 over all labels) and on every row's split between retracted and the rest (kappa 1.0). No disagreement involves retracted. Twice the first labeller found not_retracted where the second could not determine the status, and once the first read a concern where the second read a notice. Under the second labels those two rows would leave the denominator and precision would rise slightly, so the first labels, which Table 3 reports, are the more conservative.

4.5 Two secondary observations

Publisher batches set yearly medians. Across 67,845 dated notices the median time to retraction is 493 days (interquartile range 156 to 1,074). But one publisher's batch can set a year's median. IEEE conference retractions make up 4,410 of 5,011 notices in 2010 and 4,091 of 4,937 in 2011, with medians of 54 and 34 days; without IEEE those years sit at 548 and 553 days. In 2023, Hindawi's 9,673 notices have a median of 467 days against 954 for everyone else (Figure 3). A yearly trend in retraction speed is largely a trend in which publishers retracted in bulk that year.

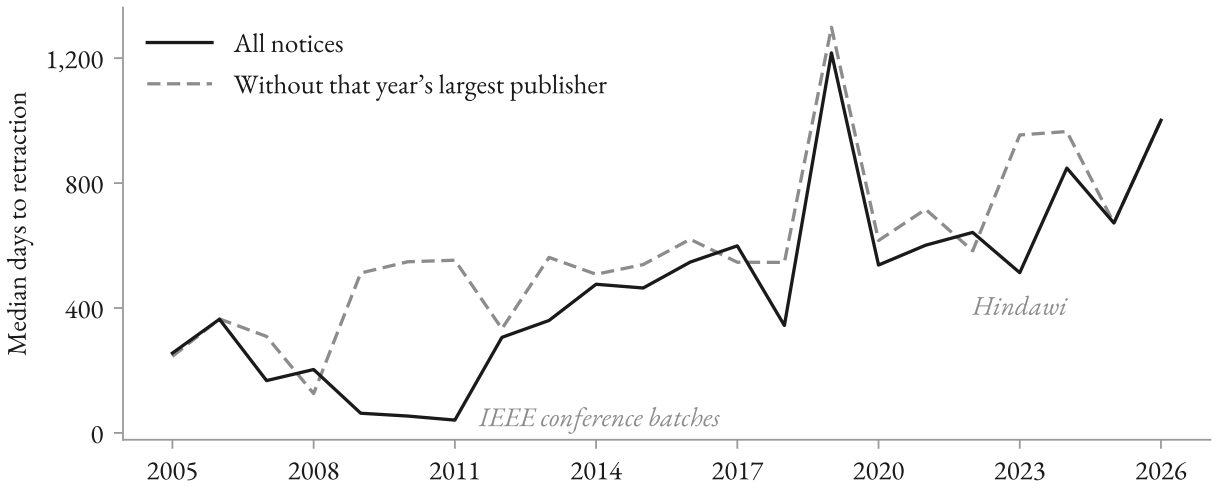


Figure 3. Median days from publication to retraction by notice year.

Retracted papers keep being cited. Of 54,737 retracted papers whose whole post retraction period falls inside OpenAlex's citation window, 32,848 (60.0%) were cited in at least one year after their retraction year, 241,815 citations in all; the median paper received 1 and the 90th percentile 10. Year granularity undercounts, and these counts do not distinguish citations that acknowledge the retraction from those that do not.

5. Discussion

The headline numbers in Section 4.1 are not errors in Retraction Watch. Late cataloguing is what a curated database does when a publisher retracts eight thousand papers at once, and renaming a reason is what a maintained taxonomy does. The difficulty lies with the studies that use it. A count from this database is a count as of a date, so a study that reports neither the date nor the file cannot be repeated.

Si et al. (2026) suggest assessing data quality over time. This paper does that for Retraction Watch, and agrees with their conclusion that researchers "may obtain different results about retraction status depending on when the data are collected". It adds a measure of how much the results differ and a split of that difference into causes.

The pattern has consequences for work already published. Counts for recent years are the most fragile: the record of 2023 grew by about half within two years, so in the current snapshot the drop from 13,336 notices dated 2023 to 6,501 dated 2024 should not yet be read as a decline. Reason based series are fragile in a second way, because a relabel moves a whole category at once on every year of the series. Studies that combine registries inherit both problems from each source, each on its own update schedule. Derived data are not exempt either: one identity rule in this study's own pipeline, corrected between preprint versions, moved 13,891 papers out of the single registry group. For systems that act on retraction status, such as reference managers that warn authors and retrieval systems that filter citations, the safe default is to treat absence from a registry as unknown, not as evidence that a paper stands.

Six practices would make retraction studies easier to repeat.

1. **Cite the snapshot.** Report the snapshot date and the Crossref commit that produced the file, not "Retraction Watch, accessed in 2025".
2. **Archive the file.** The CSV is about 68 MB; deposit the exact copy used.
3. **Match reasons on a versioned taxonomy.** Map old labels to new before counting, or a query for "Fake

Peer Review" returns zero after January 2026.

4. **Treat recent years as provisional.** In September 2024 the database held about two thirds of the 2023 retractions it held in October 2026. A study should report how its counts for the last two years have moved across recent snapshots.
5. **Filter by nature before counting.** Retraction Watch holds expressions of concern, corrections, and reinstatements beside retractions; counting all of them as retractions is the error behind the 2026 correction of Salami et al. (2024a).
6. **Filter notices before using OpenAlex's flag.** Two in five of the works it marks retracted are notices, and it flags nine in ten notice DOIs. PMC's rule, which flags only the target of a notice's link, shows the fix.

6. Limitations

- Snapshots are monthly before 2026-09-29, so an edit made and reverted within a month is invisible, and the history cannot be extended before 2024-09-02, where the public repository begins.
- Identity resolution is one hop. A paper linked only through a chain of mappings appears twice, which can undercount agreement but never overcounts it.
- OpenAlex and PubMed were fetched as positive sets, so "not flagged" there mixes "known and fine" with "unknown to that source".
- PMC coverage is targeted at the 43,616 articles other registries mention, not every PMC article. The notice link count covers PMC's open data only: 3,524 notices PMC holds outside it could not be read.
- Citation counts are per year and start in 2012.
- The second labeller covers 40 of the 200 audit rows, so agreement between labellers rests on a fifth of the sample, and blinding is partial (Section 3.5). Three clarifications of the codebook were made during labelling; each is logged with its date.

Correction

Version 1 of this preprint (2026-10-29) treated every DOI that Retraction Watch lists as a notice as a notice, including DOIs it also lists as the retracted paper's own. Because IEEE and some other publishers stamp the notice onto the paper's page, the filter discarded 11,852 OpenAlex, 1,220 PubMed, and 399 PMC flags on real papers. Version 2 fixes the rule (Section 3.4). All eligible registries now agree on 66.1% of flagged papers, not 51.5%; 22,510 papers are flagged by one registry, not 36,401; 39.2% of OpenAlex's retracted works are notices, not 49.3%; and the audit estimates 97.4% (93.7% to 100%), not 95.0%, on the corrected frame. The results of Section 4.1 are unchanged.

Use of AI tools

Generative AI (Anthropic's Claude) assisted with writing code, drafting and editing text, and checking cited sources against their originals. The author designed the study, labelled every audit row by hand from publishers' pages, traced every reported number to a run report, and takes full responsibility for the content.

Author contributions

Tanay Pratap Singh: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing (original draft), Writing (review and editing).

Competing interests

The author has no competing interests.

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Data availability

Code, run reports, the audit codebook, and the labelled sheet: https://github.com/TanayPratapSingh/afterlife. Findings in this paper come from runs/20261029T201514Z-findings.json; the audit sample from runs/20261027T004212Z-audit-sample.json; the audit scores from runs/20261029T201440Z-audit-score.json; agreement between labellers from runs/20261029T040003Z-audit-agree.json; notice links in PMC from runs/20261029T052743Z-notices.json. `afterlife findings` and `afterlife notices` rebuild every number from the landed files. The 39 Retraction Watch snapshots and the other source files fetched on 2026-10-19, exactly as fetched, with checksums, are archived at https://doi.org/10.17605/OSF.IO/D7GSF (Singh, 2026). The notice files fetched on 2026-10-29 (about 32 MB) are not yet in that archive.

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